Introduction to end-to-end analytics using Microsoft Fabric

Overview of Microsoft Fabric

- **Unified Platform**: Fabric is a software-as-a-service (SaaS) platform that integrates data engineering, integration, warehousing, real-time intelligence, data science, and business intelligence.
- **OneLake Architecture**: All data is stored in a single, open format (Delta-Parquet) within **OneLake**, a centralized "logical lake" that eliminates the need to copy or move data between different systems.
- **Shortcuts**: This feature allows users to virtualize and access data from external sources (like Azure, Amazon S3, or Google) within OneLake without moving or duplicating it.

Team Collaboration and Roles

- **Breaking Silos**: Fabric simplifies collaboration by providing a shared environment for different roles, including data engineers, data scientists, analysts, and business users.
- **Role-Specific Tools**:
  - **Data Engineers**: Use pipelines, notebooks, and lakehouses to ingest and transform data.
  - **Data Scientists**: Build and test machine learning models using integrated notebooks and Azure Machine Learning.
  - **Data Analysts**: Connect directly to OneLake using **Direct Lake mode** and create interactive reports in Power BI.
  - **Low-code/Citizen Developers**: Discover curated data via the **OneLake catalog** and use templates to create dashboards without advanced technical skills.

Administration and Governance

- **Enabling Fabric**: Admins (Fabric, Power Platform, or Microsoft 365) must enable the service via the **Admin portal** under **Tenant settings**.
- **Workspaces**: These serve as logical containers for managing items like reports and lakehouses, allowing for clear separation of resources and security control.
- **Centralized Governance**: OneLake is centrally governed, supporting sensitivity labels and metadata tracking through the OneLake catalog to maintain security and compliance.

AI Integration with Copilot

- **Generative AI Assistant**: **Copilot** is integrated across all Fabric workloads to enhance productivity using large language models (LLMs).
- **Workload Capabilities**:
  - **Data Factory**: Generates code for data transformations and explains complex tasks.
  - **Data Warehouse**: Converts natural language into SQL queries.
  - **Power BI**: Automatically generates reports and summaries of data.
  - **Real-Time Intelligence**: Translates natural language questions into **Kusto Query Language (KQL)**.

Questions

1. What is a key benefit of using Microsoft Fabric in data projects?
A. It allows data professionals to work independently, without collaboration.
B. It requires duplicating data across systems to ensure availability.
C. It provides a single, integrated environment for collaboration on data projects.
D. It eliminates the need for any cloud-based storage or internet connection.

2. What is the default storage format for Fabric's OneLake?
A. JSON
B. CSV
C. Delta-Parquet
D. XML

3. Which Fabric experience is used to move and transform data?
A. Data Science
B. Data Warehousing
C. Power BI
D. Data Factory

4. What is OneLake's primary role within the Microsoft Fabric architecture?
A. A relational database used exclusively for SQL queries.
B. A decentralized storage system where each team manages their own separate cloud.
C. A centralized "logical lake" that unifies data across regions and clouds without duplication.
D. A temporary cache for data before it is moved to a permanent warehouse.

5. Which feature allows you to access existing cloud data from external locations like Amazon S3 without copying it?
A. Data Mirroring
B. Shortcuts
C. Semantic Models
D. Data Pipelines

6. Which administrative role is authorized to enable Microsoft Fabric for an entire organization?
A. Data Scientist
B. Business Analyst
C. Fabric admin (formerly Power BI admin)
D. Database Contributor

7. How does the "Direct Lake" mode benefit Data Analysts using Power BI?
A. It requires data to be exported to a local CSV file before analysis.
B. It allows analysts to connect directly to OneLake, reducing the need for downstream transformations.
C. It limits the analyst to viewing only historical data from the previous year.
D. It forces all transformations to be performed in a separate SQL database first.

8. Which Copilot capability is specifically designed for the Data Warehouse and SQL Database workloads?
A. Generating 30-second music tracks for data presentations.
B. Converting natural language descriptions into SQL queries.
C. Designing physical hardware layouts for on-premises servers.
D. Automatically deleting outdated tables without user confirmation.

9. What is the purpose of "Workspaces" in Microsoft Fabric?
A. To provide a physical office space for data teams to meet.
B. To act as logical containers for organizing and managing data, reports, and assets.
C. To replace the need for an internet browser when accessing data.
D. To serve as a private storage area that cannot be shared with other team members.

10. Which Fabric tool provides advanced scripting capabilities for complex data transformations using Spark?
A. Dataflows
B. Power BI Templates
C. Notebooks
D. OneLake Catalog

11. In the context of governance, what does the OneLake catalog help users do?
A. Purchase additional cloud storage licenses.
B. Find and access curated data sources and monitor data sensitivity labels.
C. Write complex Python scripts for machine learning models.
D. Physically ship hard drives to Microsoft data centers.

12. Which workload is primarily responsible for processing, monitoring, and analyzing streaming data?
A. Real-Time Intelligence
B. Data Warehousing
C. Data Science
D. Data Integration

13. Which workspace role should be assigned to a user who only needs to view content without modifying it?
A. Admin
B. Contributor
C. Member
D. Viewer

Answer Key

1. C
2. C
3. D
4. C
5. B
6. C
7. B
8. B
9. B
10. C
11. B
12. A
13. D